



## chap 5.

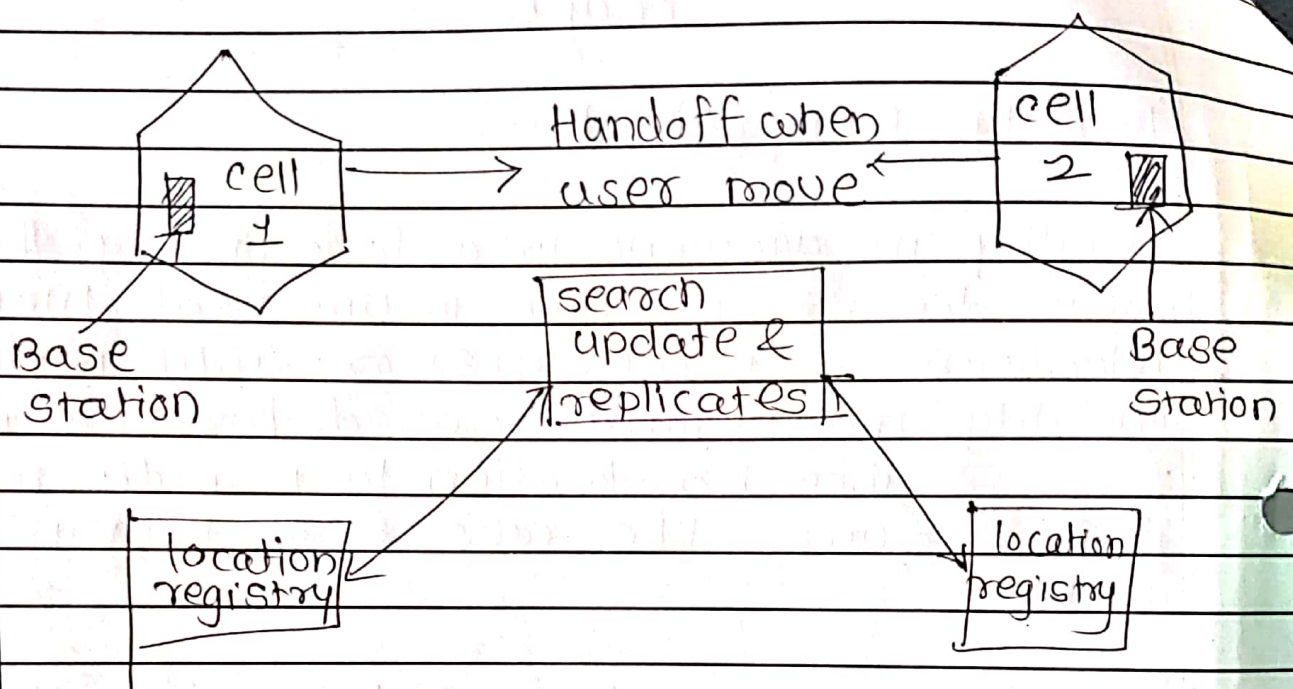
### Mobility Management -

Mobility management is a functionality that facilitates mobile devices operations in Universal Mobile Telecomm<sup>n</sup> system (UMTS) or GSM networks. mobility management is used to trace physical user & subscriber location to provide mobile phone services like calls & short msg service (SMS).

- Mobility management is nothing but the technique in which uninterrupted signal connectivity is maintained; when mobile device changes location from cell 1 to cell 2 or from n/w 1 to n/w 2.
- following are the 2 important points to ensure constant connectivity.

- 1) Infrastructure mgmt that connects two or more cells or n/w's.
- 2) location mgmt & registration mgmt by handoff when mobile devices move from one cell to another cell.

The technique of mobility management is as shown in diagram:



- Mobility mgmt is one of the major functionality of a GSM or a UMTS n/w.
- mobile devices inform the cellular n/w, whenever it moves from one location area to another.
- mobile devices detect the location area code.
- When a mobile finds that the location area code is different from its last update, it performs another update by sending to the n/w a location update request, together with its previous location, & its Temporary Mobile Subscriber Identity (TMSI) as well.
- Thus a subscriber enjoys an uninterrupted access to the n/w.



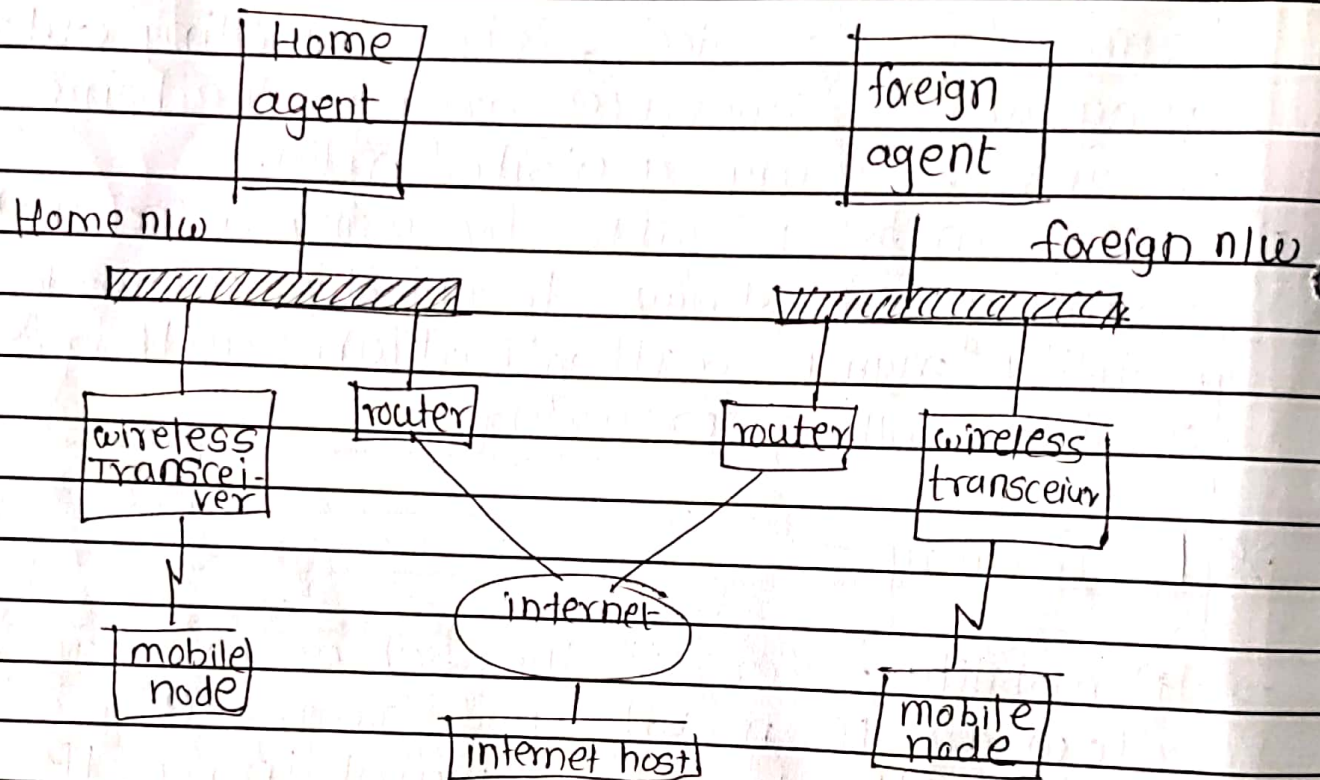
- Roaming is the fundamental mobility mgmt procedures of all cellular networks.
- Roaming is referred as the ability for a customer to automatically make & receive voice calls, send & receive data or access other services including home data services, when travelling outside the geographical coverage area of the home n/w, by means of using a visited n/w.
- This can be possible by using a comm<sup>n</sup> terminal. Roaming is always technically supported by mobility mgmt, authentication, authorization as well as billing procedures.

### \* IP mobility -

- IP mobility refers to the set of mechanisms that allow an IP mobile node to move freely bet<sup>n</sup> different IP n/w while maintaining IP connectivity in a transparent way.
- One of the most desirable features of IP mobility mechanisms is the ability of maintaining connectivity without interrupting ongoing comm<sup>n</sup>.
- i> Mobile IP ⇒
  - This is an IETF (Internet Engineering Task Force) standard comm<sup>n</sup> protocol designed to allow mobile devices users to move from one n/w to another while maintaining their permanent IP (IP) address.



→ mobile IP is an enhancement of the IP that adds mechanisms for forwarding internet traffic to mobile device when they are connecting through other than their home n/w.



- First of all, the internet host sends a datagram to the mobile node using the mobile node home addr.
- If the mobile node (MN) is on it's home n/w, the datagram is delivered through the normal IP process to the mobile node. Otherwise the Home agent picks up the datagram.
- If the mobile node (MN) is on foreign n/w the Home agent (HA) forwards the datagram to the foreign agent.
- The foreign agent delivers the datagram



to the mobile node.

→ The care-of-address (CoA) is used only for comm<sup>n</sup> with mobility agents & is never seen by the internet host.

→ Components of a mobile IP n/w -

Mobile IP has 3 major components as mentioned below -

i) Mobile Node -

A device such as a cell phone, personal digital assistant or laptop whose s/w enables n/w roaming capabilities.

ii) The Home Agent -

A router on the home n/w serving as the anchor point for comm<sup>n</sup> with the mobile node, its tunnel packet from a device on the internet called a correspondent node, to the roaming mobile node.

iii) The Foreign Agent -

A router that may fun<sup>n</sup> as the point of attachment for the mobile node when it roams to a foreign n/w delivers packets from the home agent to the mobile node.

The Mobile IP process has 3 main phases -

Phase I -





## (1) Agent Discovery -

During the agent discovery phase HA & FA advertise their services on the nlw by using the ICMP router discovery protocol (IRDP).

Mobile IP defines two methods -

### 1) Agent advertisement -

- FA & HA advertise their presence periodically using special agent advertisement msg.
- These msg advertisement can be seen as a beacon broadcast into the subnet.
- for this advertisement internet control msg protocol (ICMP) messages according to RFC 1256, are used with some mobility extension.

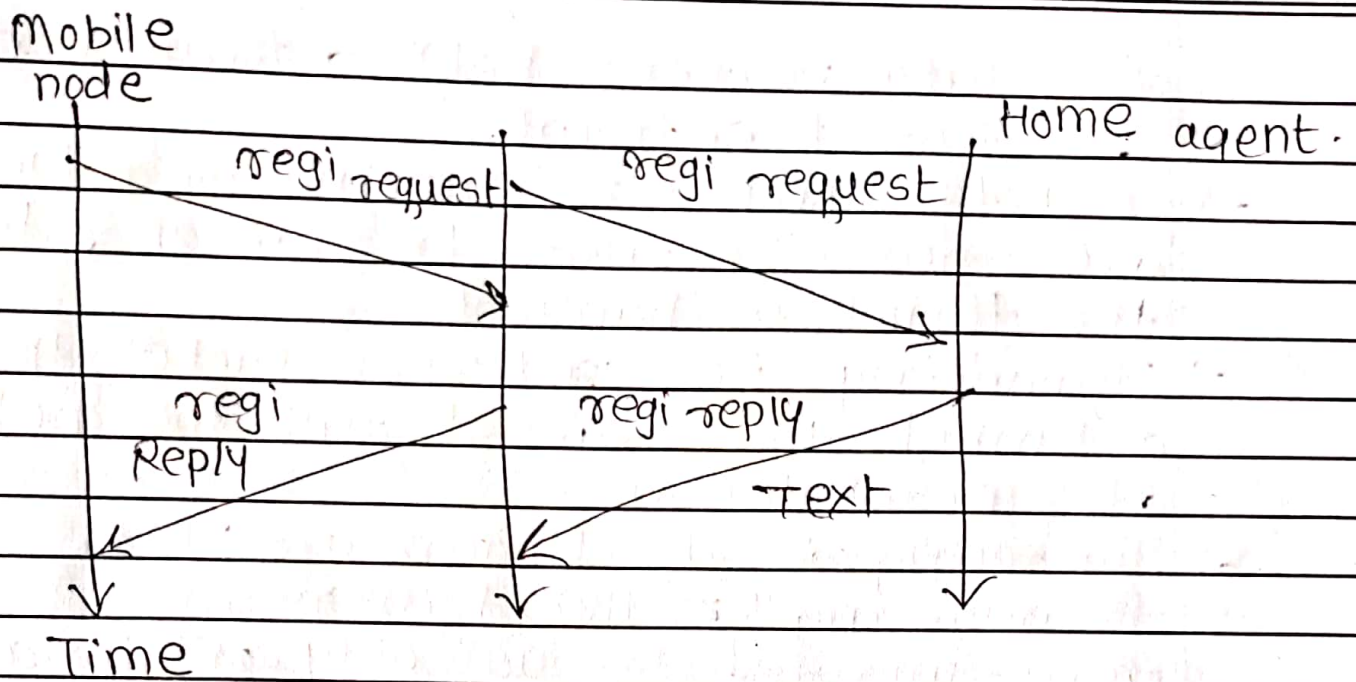
### 2) Agent solicitation -

- If no agent advertisements are present or the inter arrival time is too high & an MN has not received a CoA, the mobile node must send agent solicitations.

## (2) Registration -

The main purpose of registration is to inform the home agent of the current location for correct forwarding of packets.





→ The registration can be done in two ways depending on the location of COA.

i) If the COA is at FA, the MN sends its reg request containing the COA to the FA which is forwarding the request to the HA.

• The HA now set up a mobility binding containing the mobile nodes home IP address the current COA.

ii) If the COA is co-located, the registration can be very simpler. The mobile node may send the request directly to the HA & vice versa. This by the way is also the reg procedure for MNs returning to their home n/w.

### (3) Tunneling -

→ Tunneling is used to establish a virtual pipe



for data packets bet<sup>n</sup> a tunnel entry & a tunnel endpoint.

- Packets which are entering in a tunnel are forwarded inside the tunnel & leave the tunnel unchanged.
- Tunneling i.e sending a packet through a tunnel is achieved with the help of encapsulation.
- Tunneling is also known as "Port forwarding" is the transmission of data intended for use only within a private, usually corporate n/w through a public n/w.

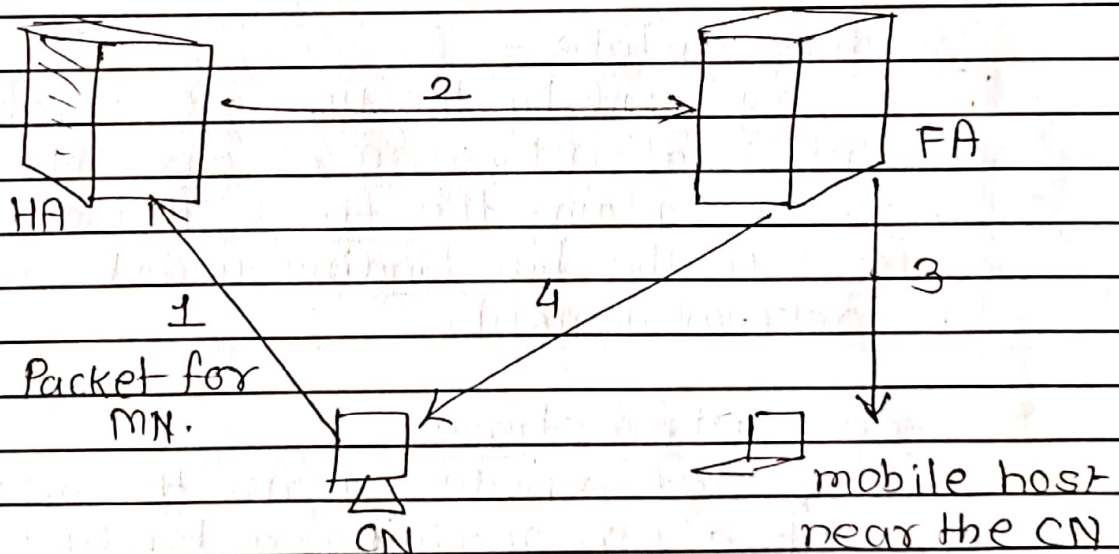


### \* Optimization -

→ If the corresponding Node (CN) & MN are very near, then also the IP packet has to travel a long way to reach the MN. This is inefficient behavior of a non optimized mobile IP called is Triangular Routing.

→ The triangle is made of the 3 segments:

- i) CN to HA
- ii) HA to CoA / MN
- iii) MN back to CN



- To solve triangular routing problem, a route optimization protocol has been introduced.
- Basically this protocol defines some messages as to inform CN of an upto date location of MN.
- Once the current location of the MN is known, the CN itself problems tunneling & sends packet directly to MN.



The optimized mobile IP protocol needs 4 additional messages: these are,

1) Binding request -

- Any node that wants to know the current location of an MN can send a binding request to the HA.
- The HA can check if the MN has allowed dissemination of its current location. If the HA is allowed to reveal the location it sends back a binding update.

2) Binding update -

- This msg sent by the HA to CNs reveals the current location of an MN.
- The msg contains the fixed IP addr of the MN & the COA. The CN binding update can request an acknowledgment.

3) Binding acknowledgement -

- If requested, a node returns this acknowledgement after receiving a binding update message.

4) Binding Warning -

- If a node decapsulates a packet for an MN but it is not the current FA for this MN, this node sends a binding warning.
- The warning contains MN's home addr & a target node addr i.e. the addr of the node that has tried to send the packet to this MN.



- The recipient of the warning then knows that the target node ~~we~~ could benefit from obtaining a fresh binding for the MN.
- The recipient can be the HA, so the HA should now send a binding update to the node that obviously has a wrong CoA for the MN.

following fig shows how these 4 msg are used together when an MN changes its FA & also shows the exchange of msg in optimization protocol.

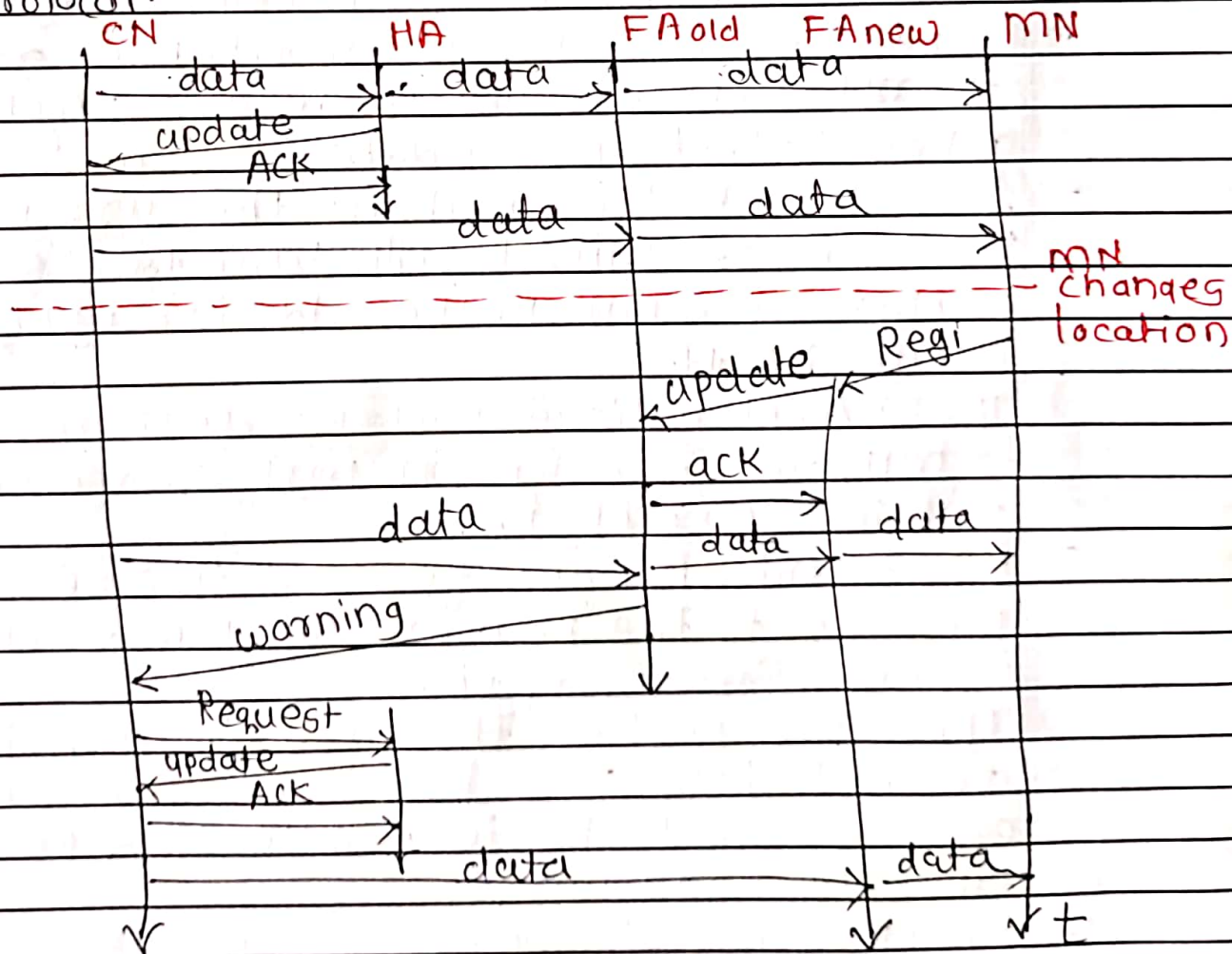


fig : Optimized mobile IP working.

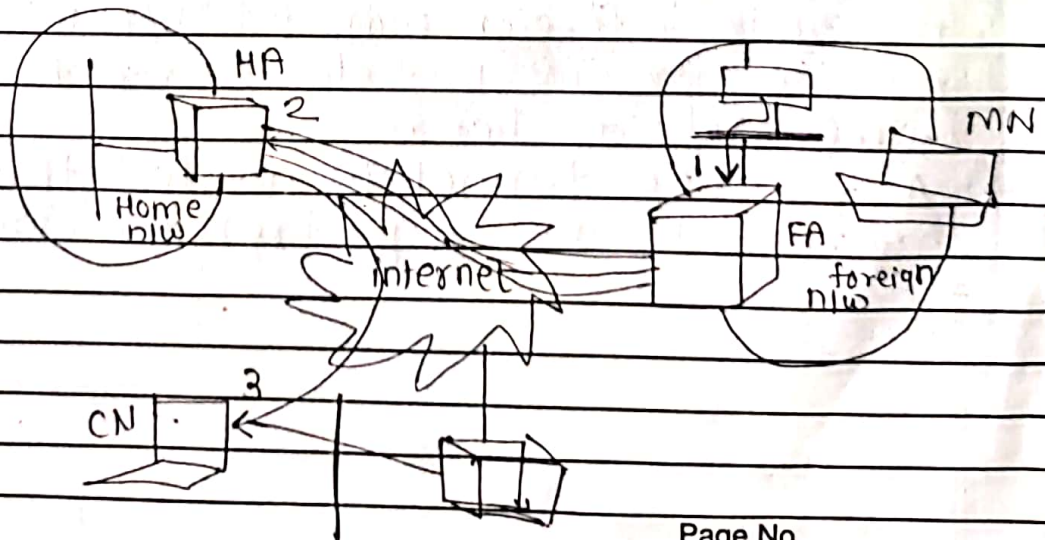
- The CN requests the current location of MN from the HA.
- HA returns the CoA of the MN via update message.
- CN acknowledges this updated message & stores mobility binding.
- Now CN can send data directly to the current foreign agent FAdd. FAdd now forwards these data to MN.
- The MN might now change its location & register with a new foreign agent FANew.
- FANew informs FAdd about new registration of MN via an update message & FAdd acknowledges this update message.
- CN doesn't know about the current location of MN, it still tunnels its packets for MN to the old foreign agent FAdd.
- The FAdd notices packets destined to MN but also knows MN currently is not in current FA.
- FAdd might now forward these packets to the new CoA of MN which is new foreign agent.
- Thus the packets that are in transit are not lost. This behaviour is another optimization to basic mobile IP & provides smooth handover.
- FAdd sends binding warning message to CN. CN then requests a binding update.



- The HA sends an update to inform the CN about the new location, which is acknowledged. Now, CN can send data directly to FA new, & avoid triangular binding.
- However, the optimization will not work if the MN does not reveal its current location to the CN bcoz of security.

### Reverse Tunnelling -

- There may be a situation where it is not feasible or desired to have the mobile node (MN) send packets directly to the internetwork via FA.
- In that case, an optional feature called reverse tunnelling is used if it is supported by mobile node, home agent & foreign agent.
- A reverse tunnel is setup betn MN & HA or betn FA & HA.
- All transmission from MN are now tunneled back to the home n/w where HA transmits them over the internet.





Reverse tunneling is used in following scenario.

1) Ingress Filtering / Firewalls -

- If the n/w where mobile node located has implemented certain security measures that prohibits the node from sending packets using its normal IP addr.
- With a reverse tunnel the packet is first encapsulated by FA & sent to the HA.

2) Multi-cast :

- Reverse tunneling is required for multicasting where the nodes in multicast group are in the home n/w, as an MN in a foreign n/w cannot transmit multicast packets directly in this case as the foreign n/w might not provide the technical infrastructure for multicast comm<sup>n</sup>.

3) TTL (Time to live) -

- When an MN is in home n/w, & if MN node moves to a foreign n/w, its TTL might be too low for the packets to reach the same node as before.
- A reverse tunnel is needed that represents one hop transmission.



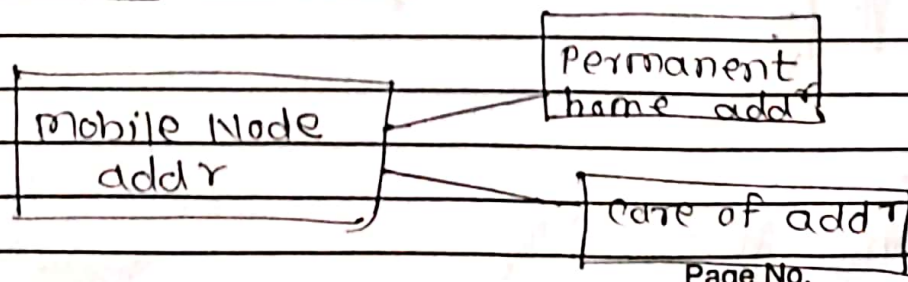
## \* IPv6 -

- Mobile IPv6 is intended to enable IPv6 nodes to move from one IP subnet to another.
- While a mobile node is away from home:
  - i) It sends information about its current location to a home agent.
  - ii) The home agent intercepts packets addressed to the mobile node & tunnels them to the mobile nodes present location.

## Features of IPv6 -

- 1) Address auto configuration -
  - Stateless auto configuration.
  - Stateful auto configuration.
- 2) Neighbor discovery -
  - Discovery each others presence & find routers
  - determine each other link layer addr.
  - maintain reachability information.
- 3) Extension headers -
  - Routing header.
  - dest<sup>n</sup> option header.

## Operational Principle -



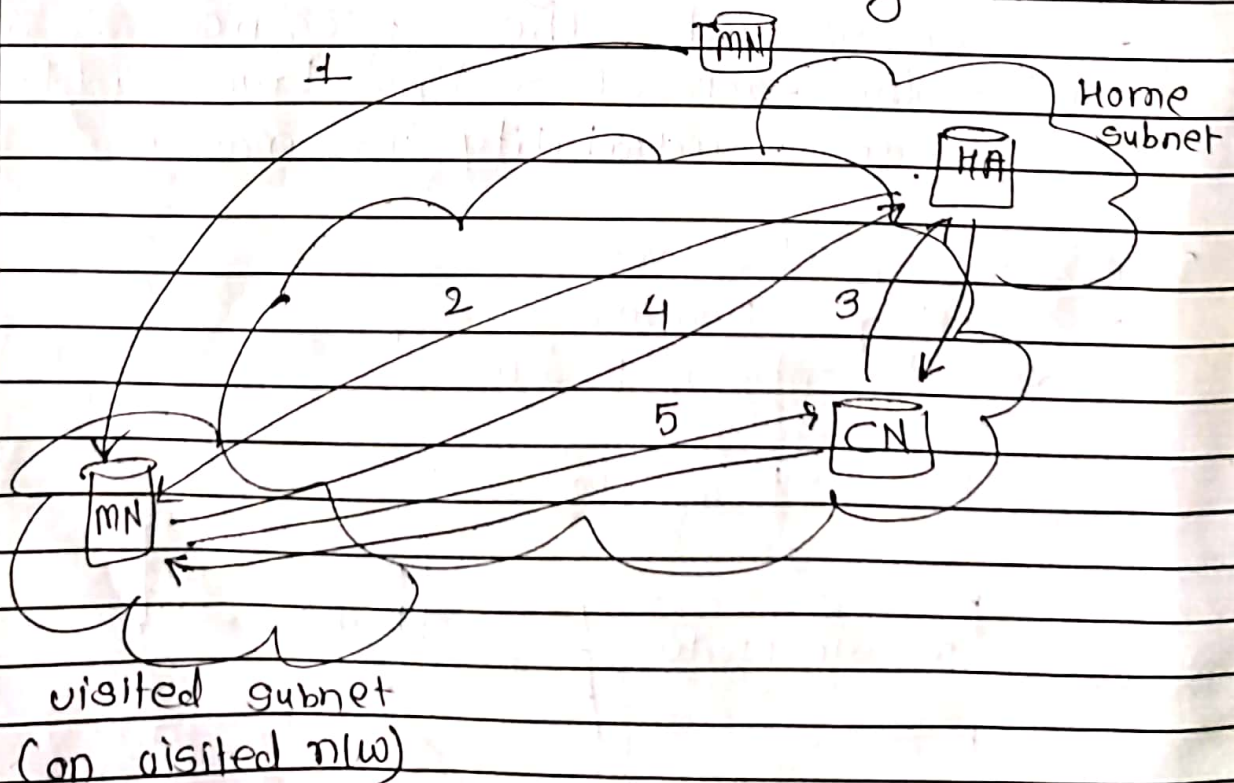


→ The goal of IP mobility is to maintain the TCP conn<sup>n</sup> bet<sup>n</sup> a mobile host & a static host while reducing the effects of location changes while the mobile host is moving around, without having to change the underlying TCP/IP.

\* Two kinds of entities comprise a mobile IP implementation.

→ A Home agent (HA) stores information about mobile nodes whose permanent home addr<sup>r</sup> is in the home agent's n/w.

→ A foreign agent (FA) stores information about mobile nodes visiting its n/w. Foreign agent also advertises care-of-addresses, which are used by mobile IP.





## Advantages & disadvantages of IPV6.

### \* Advantages -

- i) Support for security using (IPsec) internet protocol security.
- ii) More powerful internet.
- iii) Address allocation is done by the device itself.

### \* Disadvantages -

- i) It will be much harder to remember IP addr.
- ii) IPV6 is not available to m/c that run IPV4
- iii) Time to convert over to IPV6.

## Appln of IPV6 -

- 1) FTP client
- 2) Telnet client
- 3) Internet explore.



\* Macro mobility - i.e. mobile nodes movement bet<sup>n</sup> different domains, to which inter domain mobility management schemes can be employed, acting as a global mobility sol<sup>n</sup> with the advantages of flexibility, robustness, & scalability.

i) MIPv6 (Mobile IPv6) -

- > Mobile IP support of both MIPv6, but IPv6 has a couple of drawbacks. The main drawback of IPv4 is addr exhaustion, making MIPv6 the future option for mobility protocol in IP nw.
- > Mobile IPv6 is a protocol developed as a subset of IPv6 to support mobility.
- > MIPv6 is an update of the mobile IP standard designed to authenticate mobile devices using IPv6 addr.
- > In traditional IP routing, IP addr represent a topology. Routing mechanisms rely on the assumption that each nw node will always have the same point of attachment to the Internet & that each nodes IP addr identifies the nw link where it is connected.
- > In this routing scheme, if you disconnect a mobile device from the internet & want to reconnect through a diff nw, you have to configure the device with new IP addr & the appropriate netmask & default router.



→ Mobile IPv6 allows a mobile node to transparently maintain connection while moving from one subnet to another.

FMIPv6 (Fast Hand over for mobile IPv6) :-

→ Mobile IPv6 (MIPv6) enables a mobile node (MN) to maintain its connectivity to the internet when moving from one access point to another.

This process is referred to as 'handover'.

→ During handover, there is a period during which the mobile node is unable to send or receive packets. This period is called 'Hand over latency'.

→ Hand over latency results from the standards handover procedure such as movement detection, new care of addr configuration, binding updates.

→ This Hand over latency is often unacceptable to real-time traffic such as Voice over IP (VoIP).

→ The fast handover for mobile IPv6 (FMIPv6) aims at reducing the long handover latency in mobile IPv6 by fast movement detection & fast binding update.

→ It uses anticipation based on layer 2 trigger information of the mobile node (MN) to obtain a new care-of-addr at the new link while still connected to the previous link, thus reducing handover delay.

→ Furthermore, it also reduces packet loss by buffering before the real link layer handover takes place.





\* Micro mobility -

- Mobile IP exhibits several problems regarding the duration of handover & the scalability of the regi procedure.
- Assuming a large no. of mobile devices changing n/w quite frequently, a high load on the home agents as well as on the n/w is generated by regi & binding update message.
- IP micro-mobility protocol can complement mobile IP by offering fast & almost seamless handover control in limited geographical areas.
- The following section presents three of the most commonly used approaches:

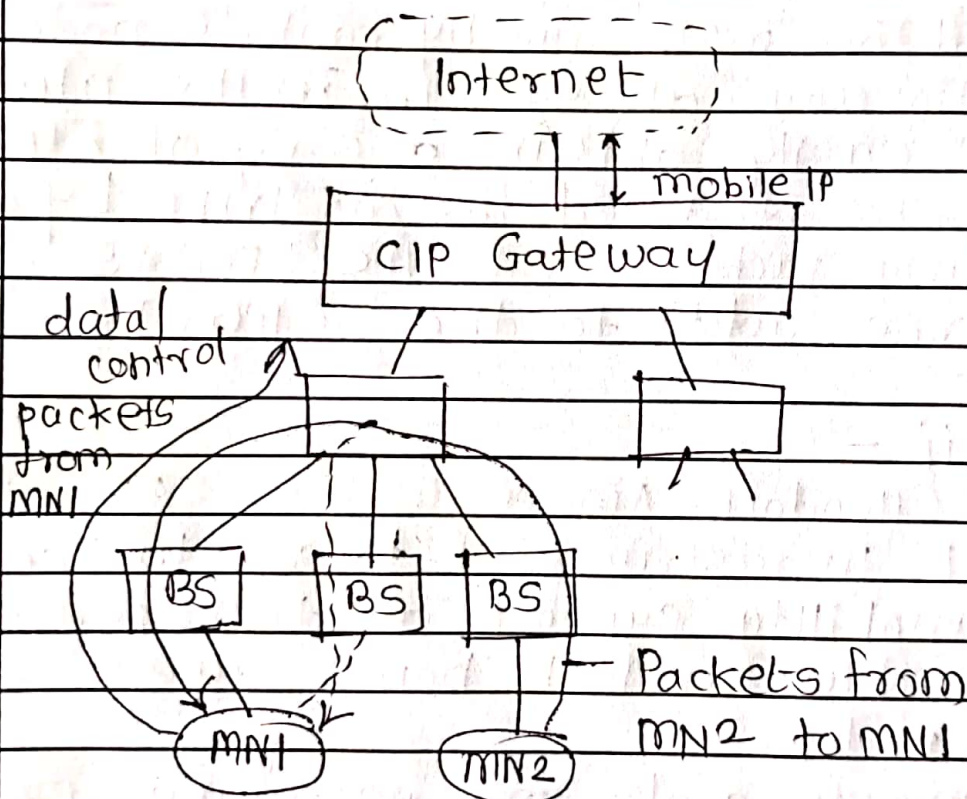
- 1) cellular IP
- 2) Hawaii
- 3) Hierarchical mobile IPv6 (HMIPv6)

1) cellular IP -

- cellular IP provides local handovers without renewed regi by installing a single cellular IP gateway (CIPGW) for each domain, which acts to the outside world as a foreign agent.
- Inside the cellular IP domain, all nodes collect routing information for accessing MNs based on the origin of packets sent by the MNs towards the CIPGW.



- Soft handover are achieved by allowing simultaneous forwarding of packets destined for a mobile node along multiple paths.
- A mobile node moving bet<sup>n</sup> adjacent cell will temporarily be able to receive packets via both old & new base station (BS) if this is supported by the lower protocol layers.



Advantage -

- 1) Manageability - cellular IP is mostly self-configuring & integration of the CIPGW into a Firewall would facilitate administration of mobility-related functionality. This is, however not explicitly specified.
- 2) cheap Passive connectivity.

3) Flexible Handoff

4) Simple memory less mobile host.

Disadvantages -

1) efficiency - Additional n/w load is induced by forwarding packets on multiple path.

2) Transparency - changes to MNs are required.

3) security - Routing tables are changed based on msg sent by mobile nodes. additionally, all system in the n/w can easily obtain a copy of all packets destined for an MN by sending packets with the MN's source addr to the CIPGW.

(2) Hawaii -

→ Hawaii (Handoff-Aware Wireless Access Internet Infrastructure) tries to keep micro-mobility support as transparent as possible for both home agent & mobile nodes.

→ It's concrete goals are performance & reliability improvements & support for quality of service mechanisms.

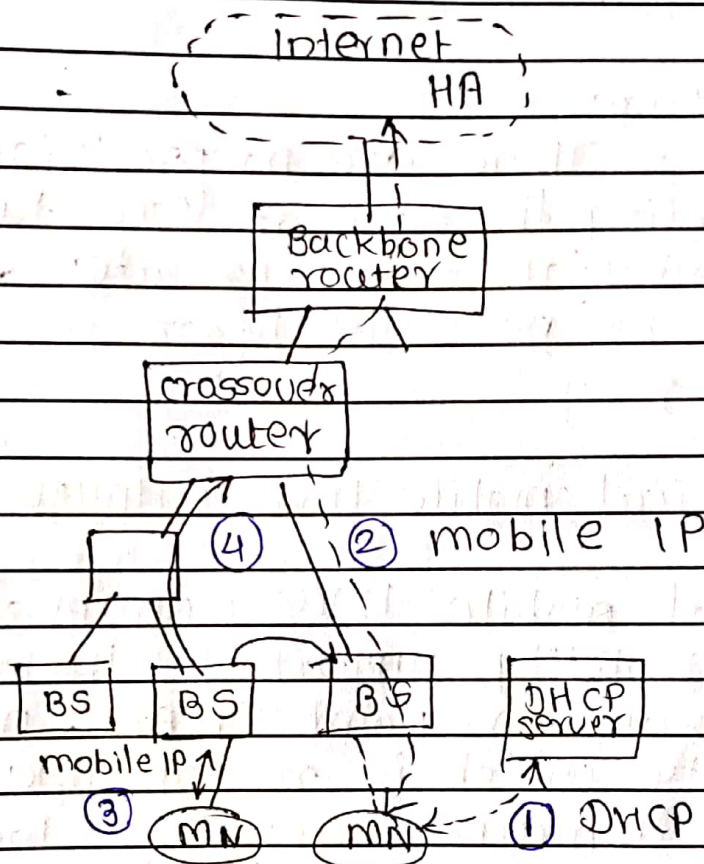
→ On entering an HAWAII domain, a mobile node obtain a co-located CoA (step 1)

→ Registers with the HA (step 2)

→ Additionally, when moving to another cell inside the foreign domain, the MN



sends a registration request to the new base station as to a foreign agent (step 3)



→ Thus mixing the concept of co-located CoA & Foreign agent CoA.

The base station intercepts the registration request & sends out a handoff update message, which reconfigures all routers on the path from the old & new base station to the so-called crossover router (step 4).

Advantages -

> security - challenge-response extension are mandatory. In contrast to cellular IP routing changes are always initiated by



the foreign domain's infrastructure.

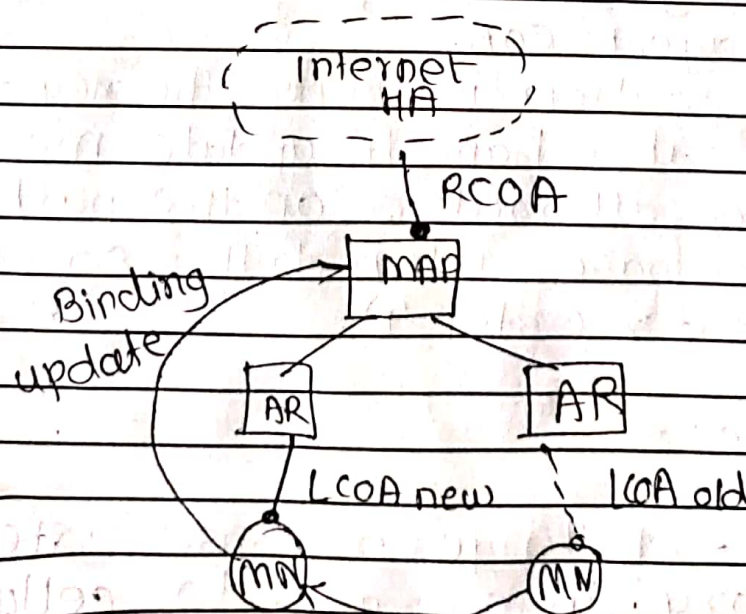
- 2) Transparency - HAWALL is mostly transparent to mobile nodes.

Disadvantages -

- 1) Security - There are no provisions regarding the setup of IPsec tunnels.
- 2) Implementation :- No private addr support is possible becoz of co-located COAs.

### Hierarchical mobile IPv6 (HmIPv6)

→ Hierarchical mobile IPv6 (HmIPv6) provides micro-mobility support by installing a mobility anchor point (MAP). MAP is an entity which is responsible for a certain domain & act as a local HA within for this domain for visiting MNs.





- The MAP receives all packets on behalf of the MN, encapsulates & forwards them directly to the MN's current addr LCOA (link COA).

### Advantages -

- 1) security - MNs can have location privacy bcoz LCOAs can be hidden.
- 2) Efficiency - direct routing bet<sup>n</sup> CNs sharing the same link is possible.

### disadvantages -

- 1) Transparency - additional infrastructure component (MAP)
- 2) security - Routing table are changed based on msg sent by mobile nodes. This requires strong authentication & protection against denial of service attacks.